

Carbon-Aware and Cost-Effective Geo-Distribution for Latency-Critical Microservices

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A. Problem Space: Carbon-Aware Microservices (MS)

Problem: How can a MS company reduce its carbon emissions and infrastructure cost?

Existing Approaches
(Only for Global Scale): **Workload shifting**

Temporal: Wait until energy is greener.



(latency critical apps: not applicable)
Spatial: Move to a greener region. (strict budgets: can be applied but carefully)

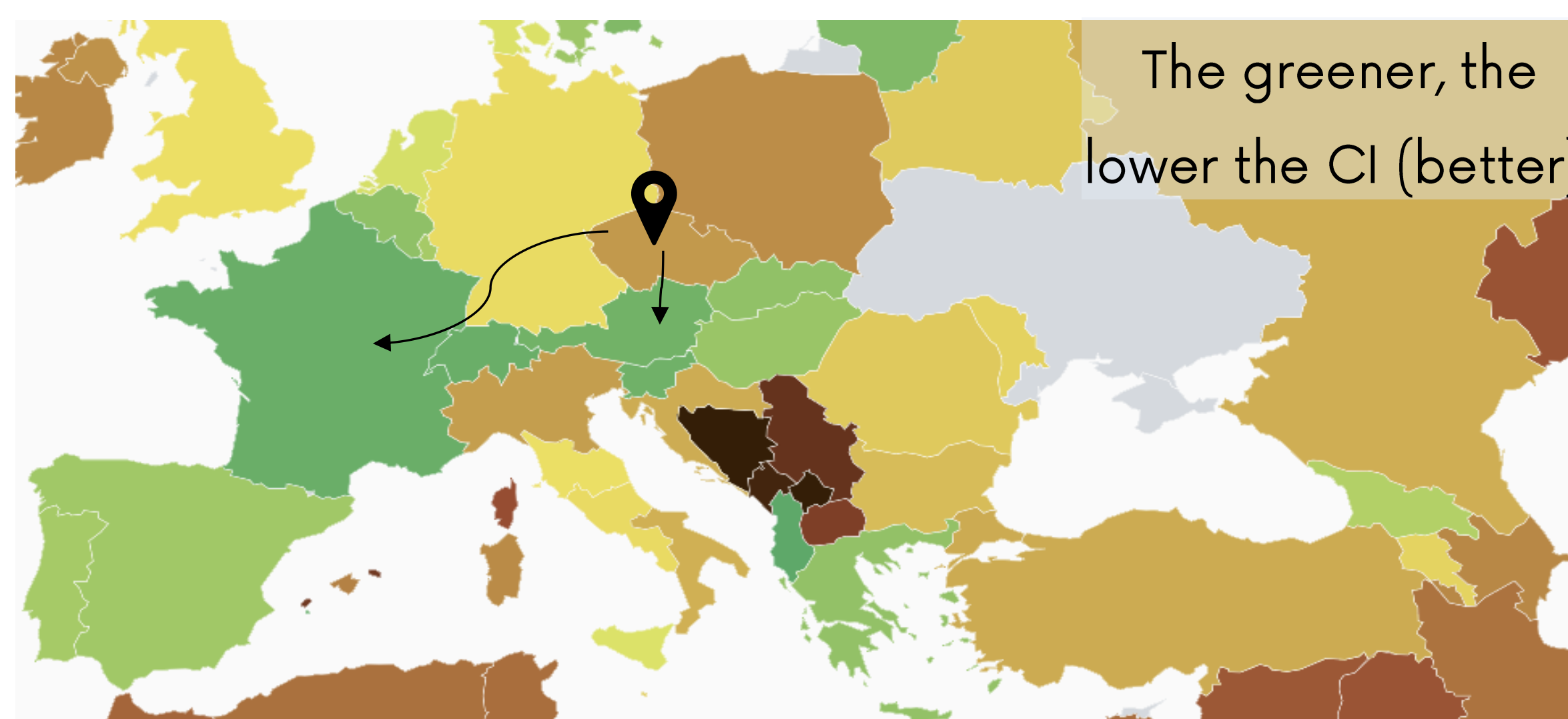


Fig. 1: Carbon intensity (CI) across different countries.

Use case: Small to medium-sized (SME) taxi reservation company in Czechia.



Harder Problem
Can a geo-constrained SME reduce both carbon and cost?

B. Insight: Move Early-Activated MS to Greener Regions

Goal: Map MS ↔ regions for

↓ Carbon ↓ Cost ⚡ Latency

Challenge: Which MS to move?
Where to move them?

⚠ Explosive Combinatorial Search Space

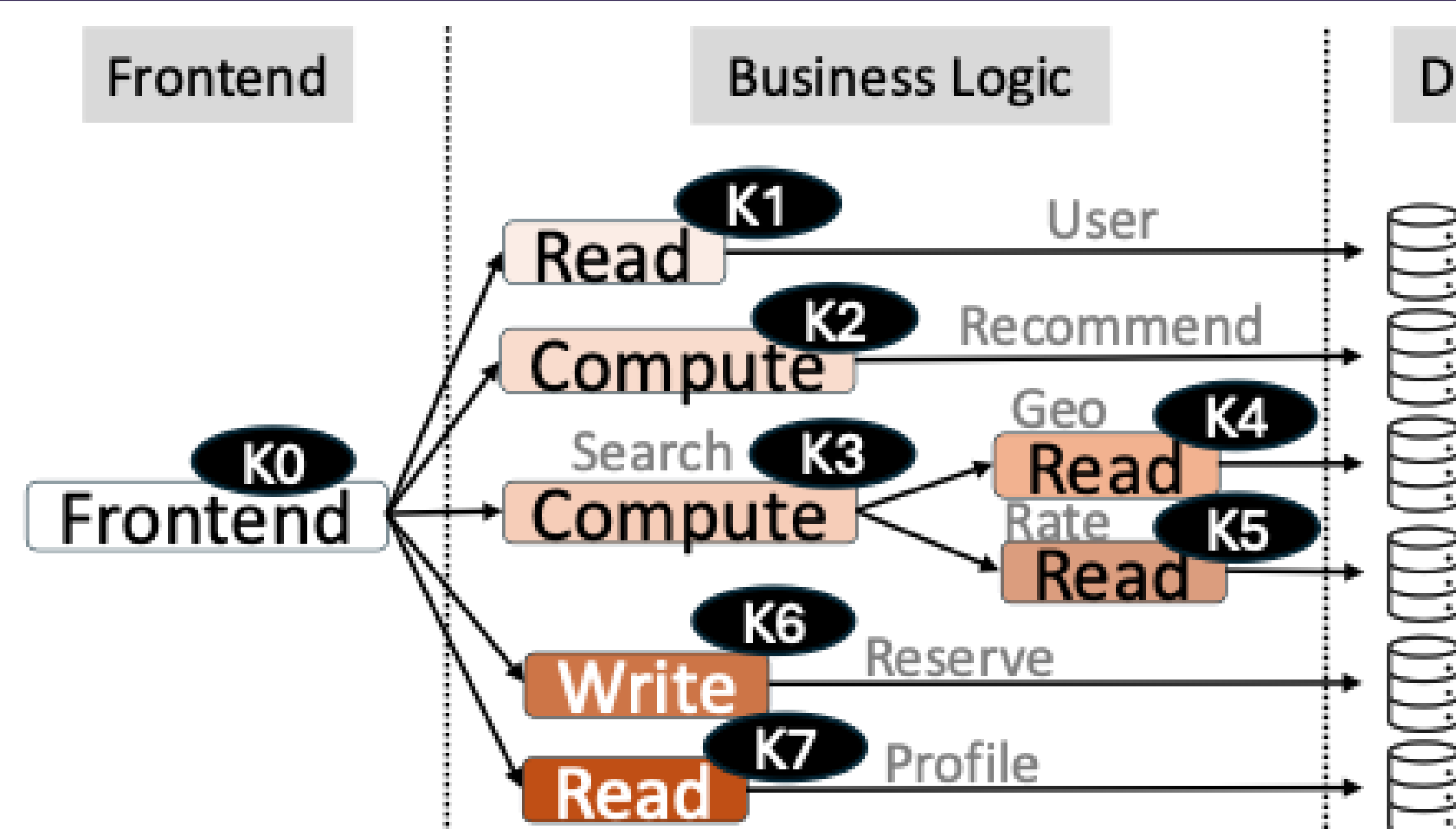


Fig. 2: MS Architecture of Hotel Reservation App.

Insight on MS Placement:

Which MS to move?

Early-activated MS subtrees

(amortize additional geographic latency over the remainder of the request execution)

Where to move them?

Strictly better regions (carbon & cost)
(converge faster & give better solutions)

C. System Design & Evaluation

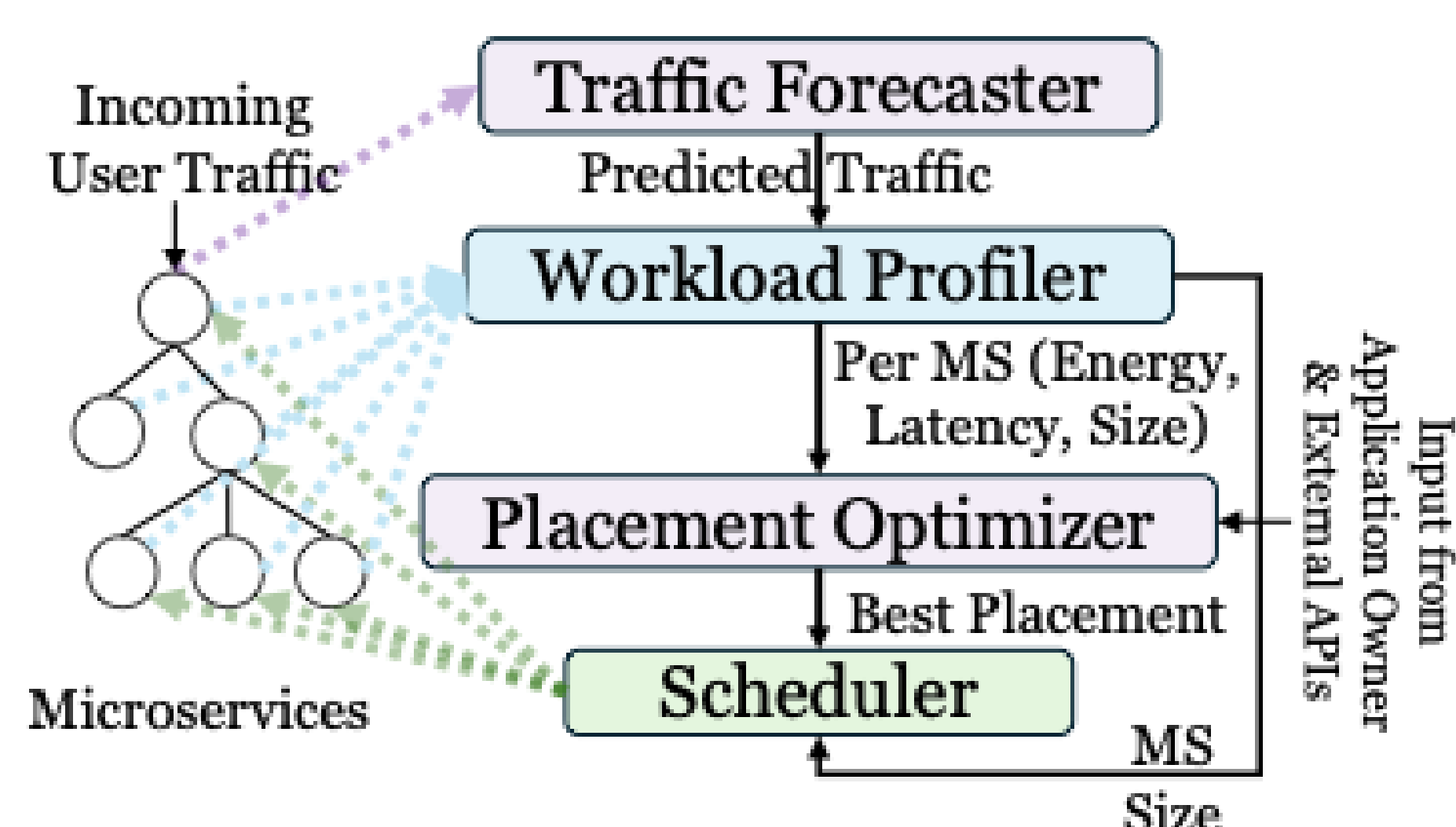


Fig. 3: System Components of Aceso.

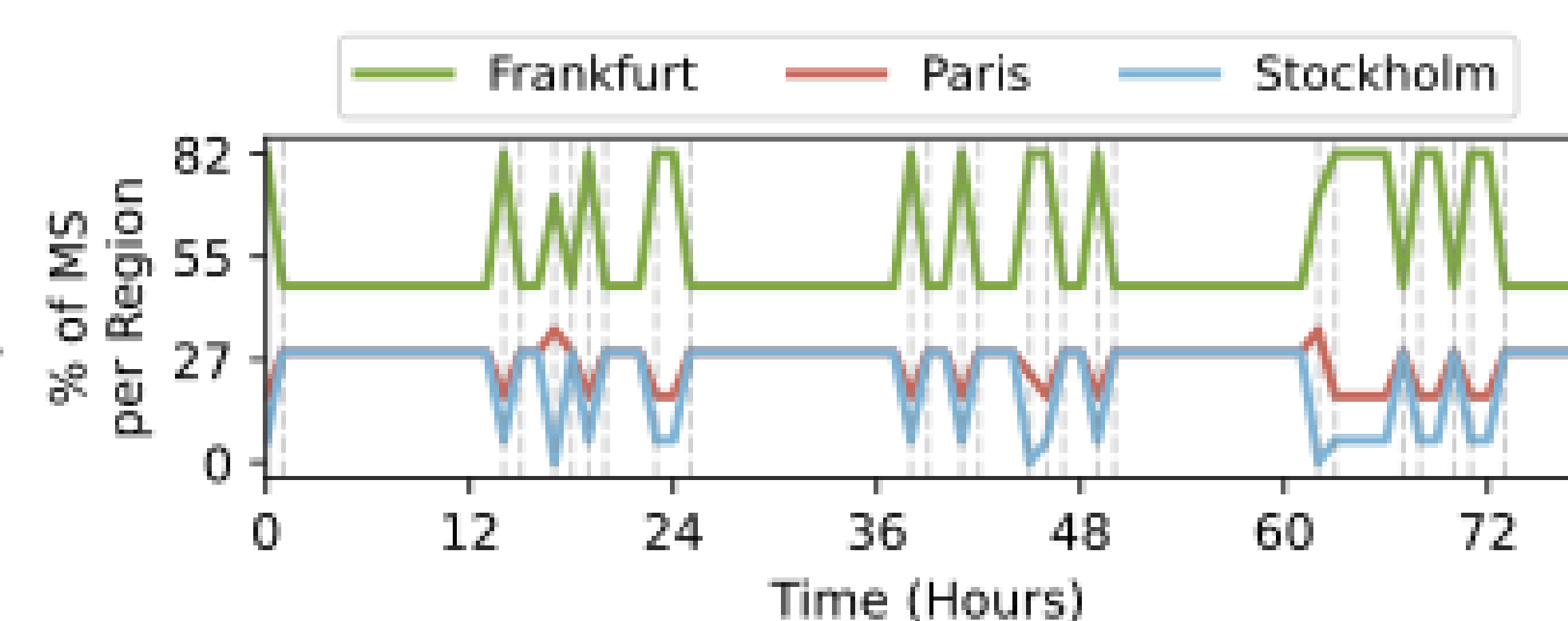


Fig. 4: MS Geo-distribution over time. Base region: Frankfurt.

Aceso solves an optimization problem to determine the map MS ↔ regions, effectively pruning the search space.

Aceso offloads 54% of MS to Paris and Stockholm and places them back to Frankfurt when traffic spikes to meet the latency goals.

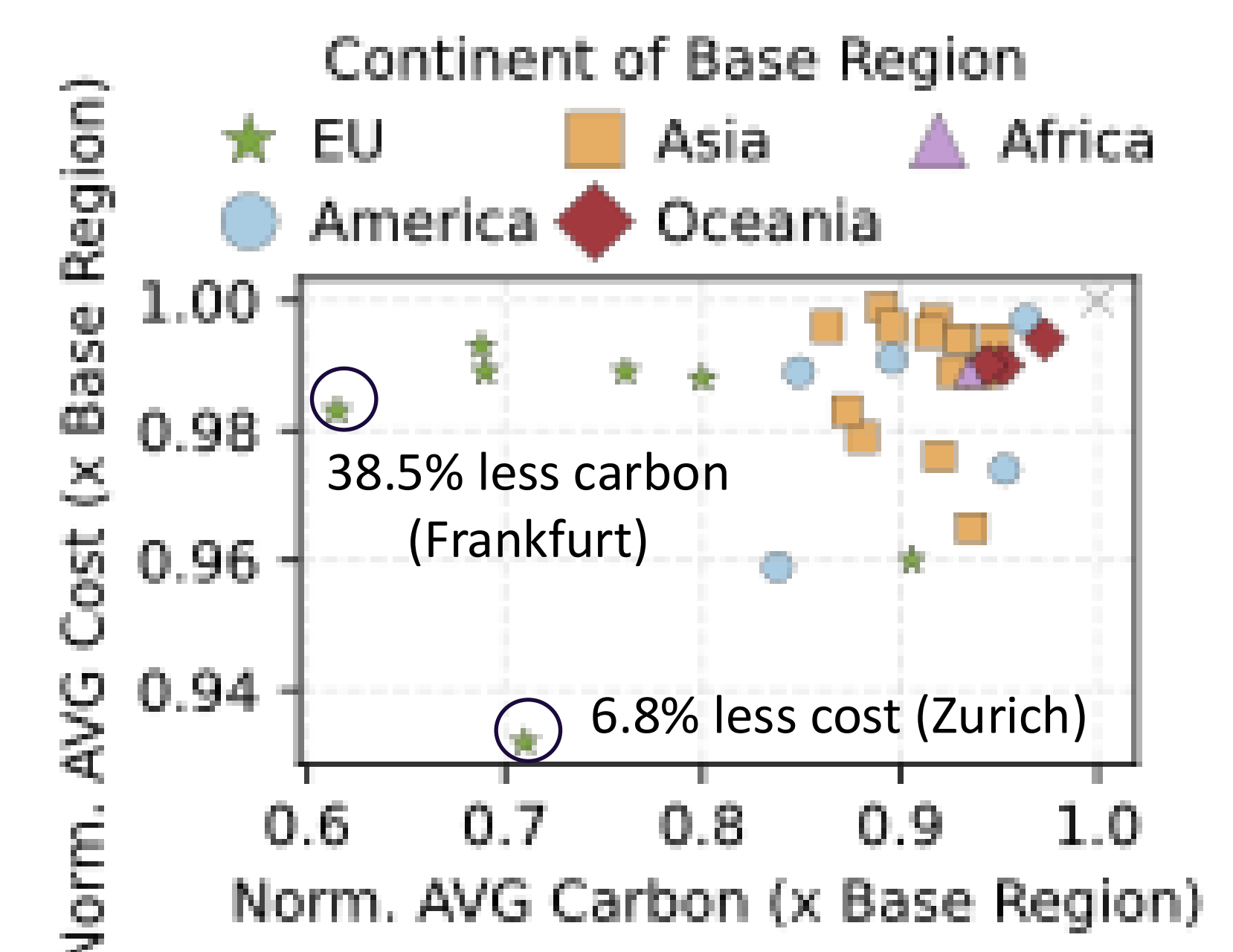


Fig. 5: Geographic Generalizability of Aceso.

EU-based SMEs achieve the largest carbon and cost reductions.

D. Comparison with Related Works

Related Work	Method
Caribou	Spatial Shifting for Serverless
Nautilus	MS Load Balancing (Single Region)
Genetic Alg. (GA)	Aceso w/o search space pruning
Linear Progr. (LP)	Exact optimization solver but very slow

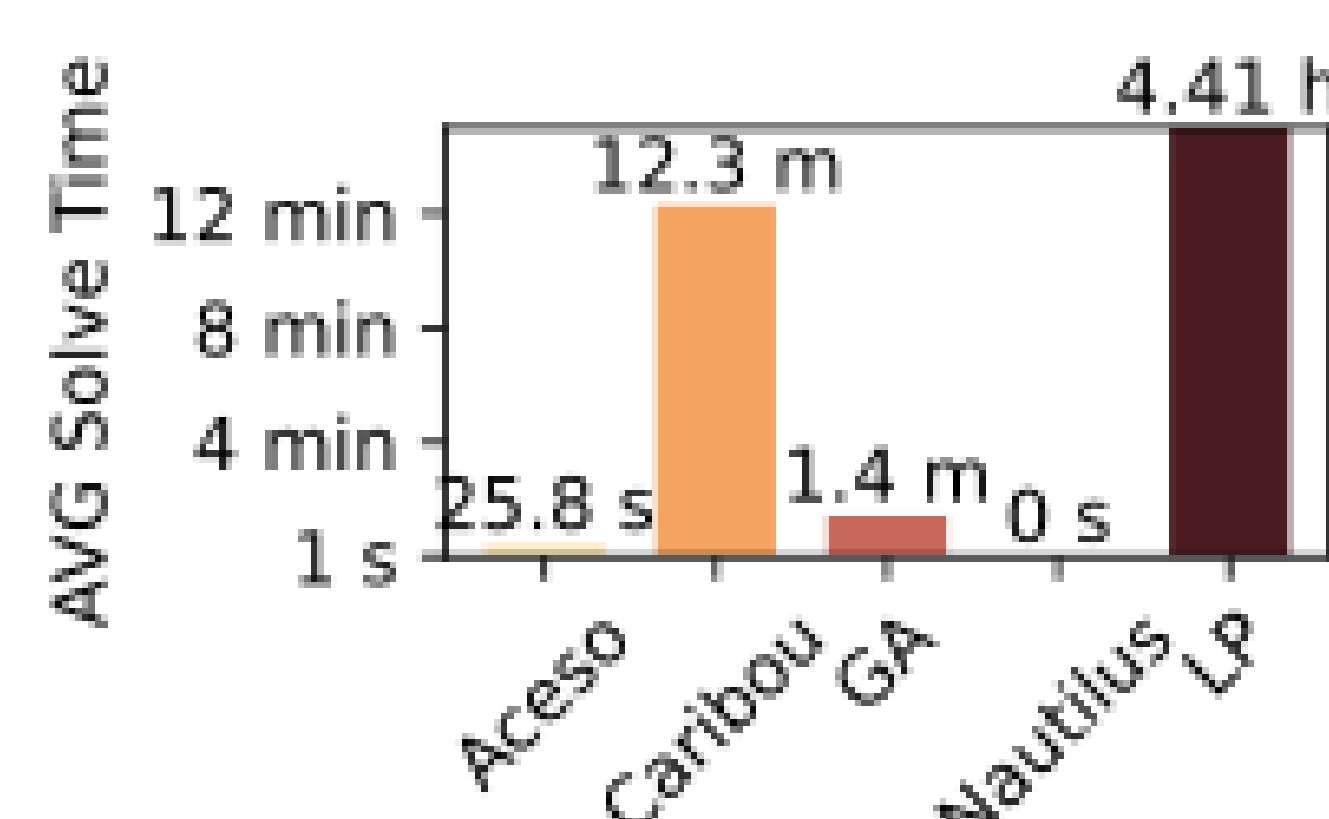


Fig. 6: Optimization solve time.

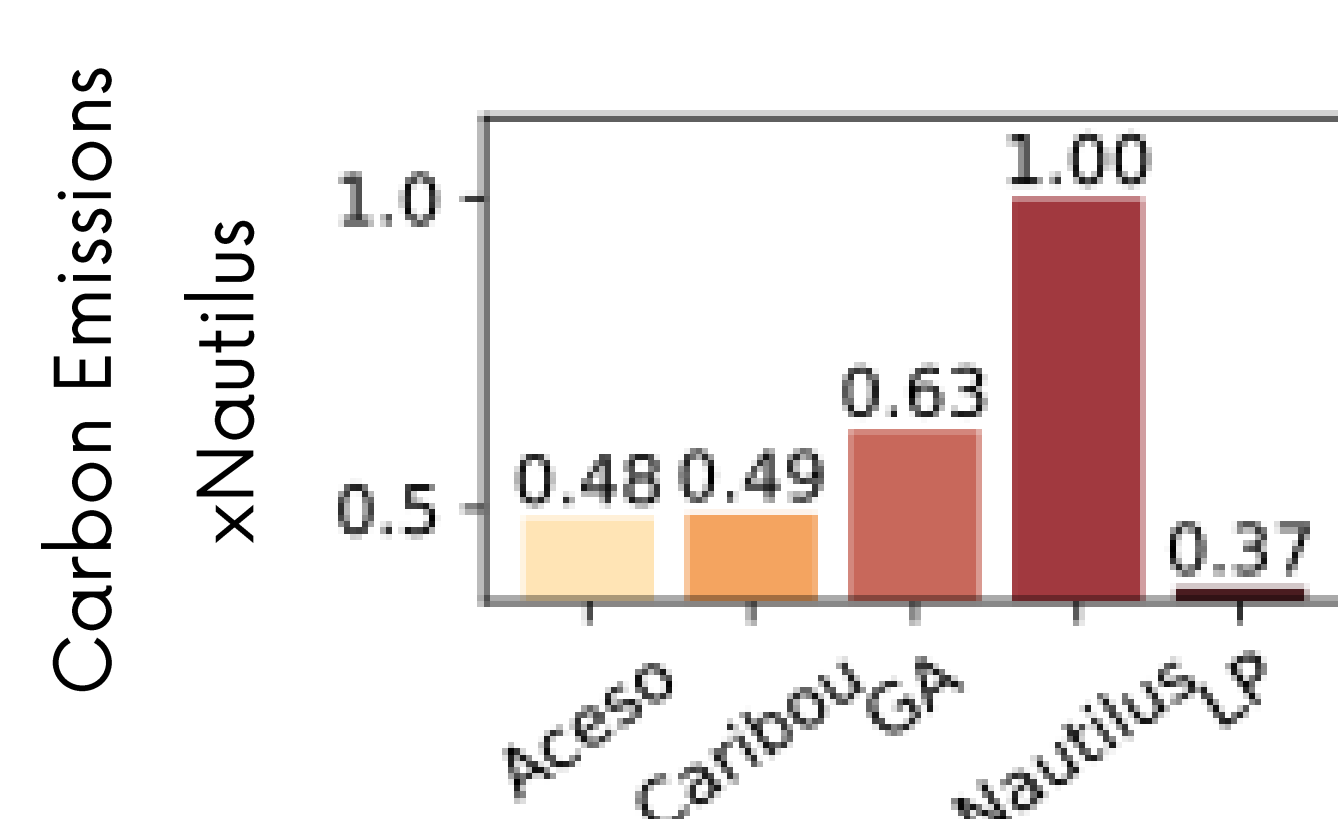


Fig. 7: Carbon Reduction (norm. Nautilus).

Takeaway: Aceso matches LP-level carbon savings while converging in seconds, far faster than other approaches, due to the use of GA and its insight-based search space pruning.

Full Paper to appear at **Middleware'26**

Aceso: Latency-Aware Geo-Distribution of Microservices for Cost-Effective Carbon Reduction

Aceso dynamically places MS across geo-constrained settings to reduce carbon emissions and cost while meeting latency constraints.

To be open sourced...

Where Does the Light Go?

A light wave is an electromagnetic wave visible to the human eye.

A.

Introduction

The behavior of light waves depends on the medium it travels through. When light travels in another medium, it can be reflected, transmitted, or absorbed.

B.

Objective

The research aims to explore which materials will reflect, absorb, and transmit light.

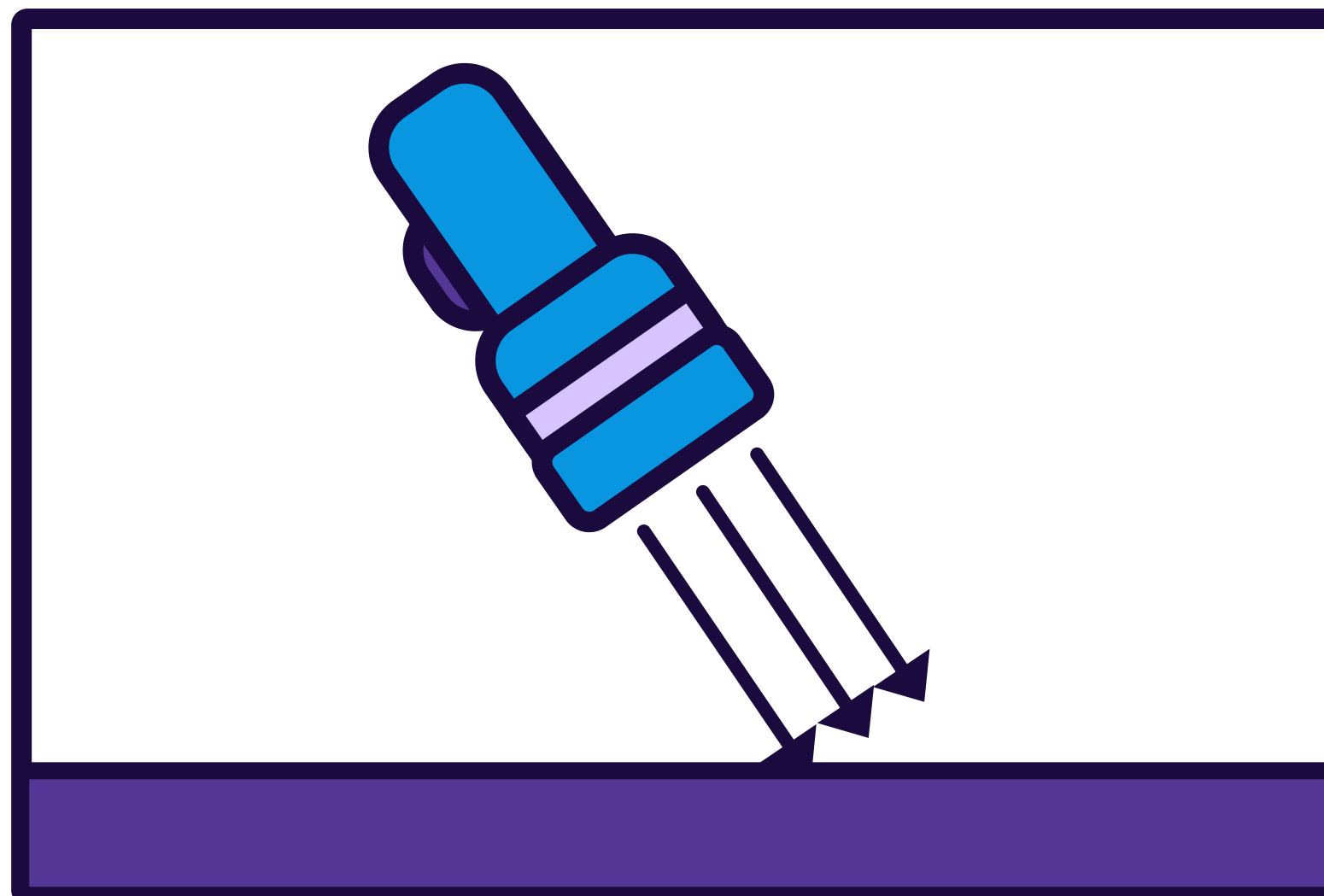
C.

Methodology

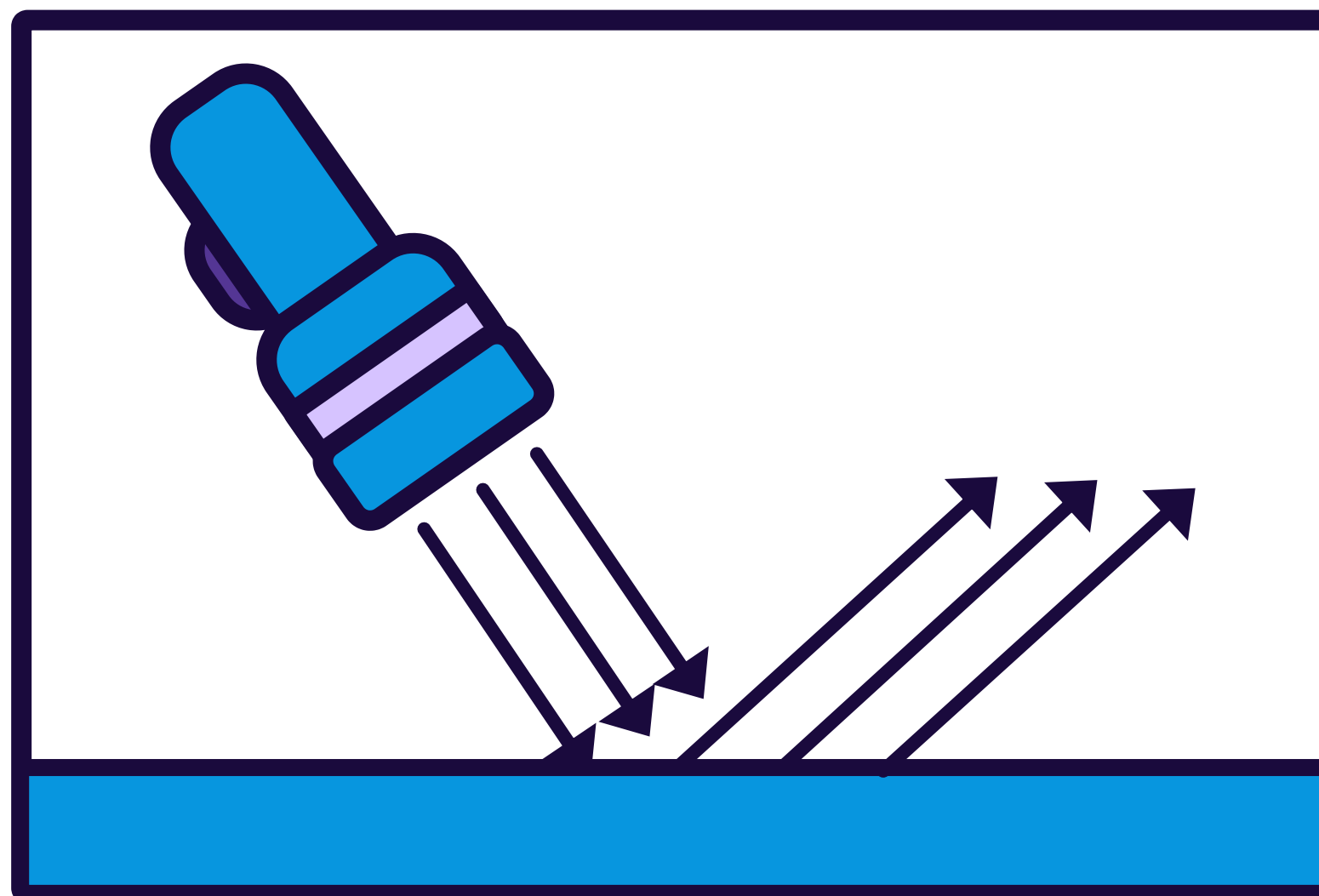
The flashlight is the source of light for this experiment. The interaction of the light wave with another medium will be observed. The media are water, mirror, and cardboard.

D.

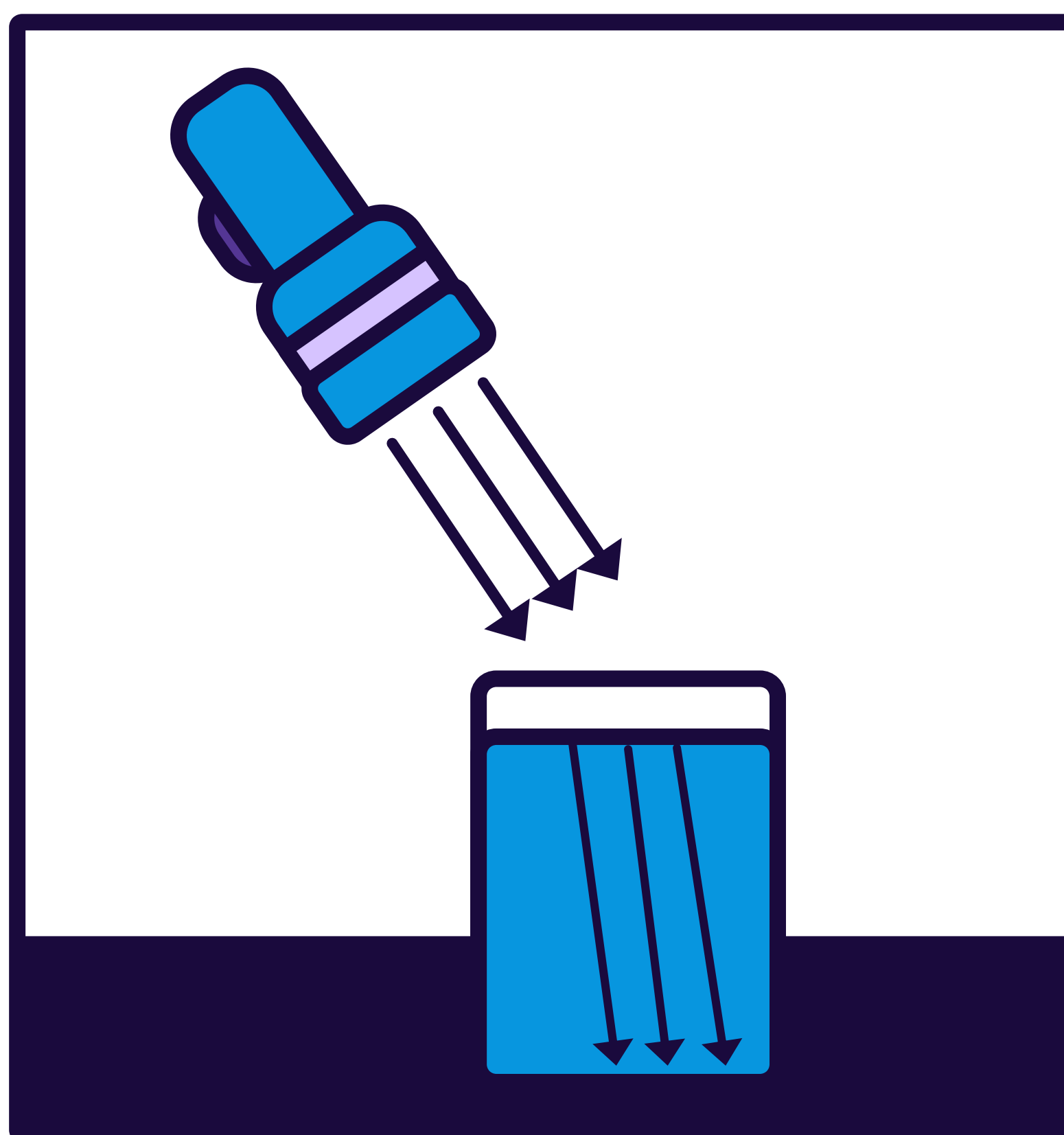
Results



When the light travels through the cardboard, the cardboard absorbs the light, hindering the light to pass through.



When the light travels through the mirror, the mirror reflects the light, at the same angle the light reached the mirror.



When the light travels through the water, the light passes through the water. The light transmits and bends through the medium.

E.

Analysis

Light will not pass through surfaces or media that are opaque. The medium will absorb the light and block it from passing through. Light will reflect on smooth and shiny objects like the mirror. The medium will bounce the light. Light, when transmitted to another medium, will bend because it will have different speeds. Water is an example of a medium where light can bend and pass through.

F.

Conclusion

The behavior of light waves depends on the medium it interacts with. The researchers recommend studying how other electromagnetic waves travel in different media.

RESOURCES

A.

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